



Islamic University of Technology
Department of Computer Science and Engineering

Lab 8: Dynamic Programming

CSE 4404: Algorithms Lab(SWE)
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Task A. Coin Combinations

Time Limit: 1 second | Memory Limit: 256 MB

A shop accepts n distinct coin denominations and keeps an unlimited supply of each. Count the number of distinct **combinations** of coins whose values sum exactly to x . Two combinations are considered the same if they use the same coins the same number of times, regardless of order, only the multiset of coins used matters. Since the answer can be large, print it modulo $10^9 + 7$.

Input Format

The first line contains two integers n and x . The second line contains n integers $c_1, c_2, \dots, c_n$ — the coin denominations.

Output Format

Print the number of distinct coin combinations summing to x , modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 100$
- $1 \leq x \leq 10^6$
- $1 \leq c_i \leq 10^6$

Examples

Sample Input	Sample Output
3 9 2 3 5	3
2 4 1 2	3

Task B. Book Shop

Time Limit: 2 seconds | Memory Limit: 512 MB

You are in a book shop which sells n different books. You know the price and number of pages of each book. You have decided that the total price of your purchases will be at most x . What is the maximum number of pages you can buy? You can buy each book at most once.

Input Format

The first line contains two integers n and x — the number of books and the maximum total price.

The next line contains n integers $h_1, h_2, \dots, h_n$ — the price of each book.

The last line contains n integers $s_1, s_2, \dots, s_n$ — the number of pages of each book.

Output Format

Print one integer: the maximum number of pages.

Constraints

- $1 \leq n \leq 1000$
- $1 \leq x \leq 10^5$
- $1 \leq h_i, s_i \leq 1000$

Examples

Sample Input	Sample Output
4 10 4 8 5 3 5 12 8 1	13
3 5 2 3 4 3 4 5	7

Task C. Register Change

Time Limit: 1 second | Memory Limit: 256 MB

A cashier's till has only cnt_i coins of denomination c_i remaining, for $i = 1, \dots, n$. A customer is due exactly x in change. Determine the minimum number of coins needed to make exactly x using the till's available stock, or determine that it is impossible.

Input Format

The first line contains two integers n and x . Each of the next n lines contains two integers c_i and cnt_i — the value of the i -th denomination and how many such coins remain.

Output Format

Print the minimum number of coins needed to make exactly x , or -1 if it cannot be done with the available coins.

Constraints

- $1 \leq n \leq 100$
- $1 \leq x \leq 1000$
- $1 \leq c_i \leq 1000$
- $1 \leq cnt_i \leq 100$

Examples

Sample Input	Sample Output
3 11 1 100 3 100 5 100	3
2 6 5 1 2 1	-1

Marks Distribution

Task	Marks
Task A	30%
Task B	35%
Task C	35%

Practice Problems

Problem	Link
Money Sums	CSES: https://cses.fi/problemset/task/1745
Minimizing Coins	CSES: https://cses.fi/problemset/task/1634
Knapsack 1	AtCoder EDU DP E: https://atcoder.jp/contests/dp/tasks/dp_e